



GED 211

Capital Budgeting

Investment & Finance

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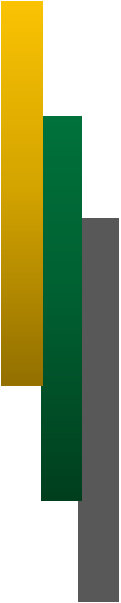
Contents: Focus of Discussion

Capital Budgeting Concepts

- Capital Budgeting Process
- Some Important Concepts of Capital Budgeting

Capital Budgeting Tools

- Payback Period (PBP)
- Net Present Value (NPV)
- Internal Rate of Return (IRR)
- Profitability Index (PI)
- Accounting Rate of Return
- Capital Rationing



Capital Budgeting

The Process

Capital Budgeting: Introduction



Capital Budgeting: Formal Definition

Capital budgeting is the planning and decision-making process through which a firm evaluates and selects long-term investment projects

- **Investment in new machinery, buildings, technology, or expansion**
 - **That will generate future cash flows & is expected to increase the firm's value**

Capital Budgeting: Definition Simplified

Capital budgeting is the process businesses use to decide which long-term projects or investments are worth spending money on

Capital Budgeting: Process



Capital Budgeting: Process

1. Identify Investment Opportunities

- The first step is to look for potential long-term projects
 - Buying new machinery
 - Launching a new product
 - Opening a new branch
 - Installing new technology

2. Plan in line with Business Goals & Strategic Fit

- Consider projects that would help the business reach its long term goals
- The project must be suitable with the rest of the business operations
 - A project might be very profitable by itself, but if it harms other parts of the business, it might not be the best choice

Capital Budgeting: Process

3. Estimate Financial Aspects

- Determine all costs (cash outflows) & All benefits (cash inflows) of each project
- Estimate:
 - Initial Investment
 - Annual cash flows
 - Terminal Cash Flow at end of project

4. Evaluate The Projects

- Capital budgeting techniques for evaluation
 - NPV (Net Present Value)
 - IRR (Internal Rate of Return)
 - Payback Period (PBP)
 - Profitability Index (PI)
 - Accounting Rate of Return (ARR)

5. Select the Best Project

- Choose the project that adds the most value to the firm
- Which in turn will maximize shareholders wealth

Capital Budgeting: Process



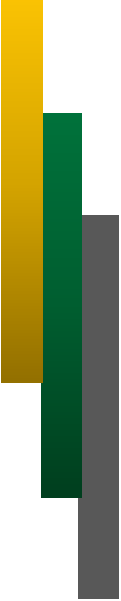
5. Implement the Selected Project

Take necessary steps to get the project running

- Purchase equipment
- Hire staff
- Construct/install the setup
- Operational setup

6. Monitor and Review Performance

Continuously monitor the projects, and in case of deviation of results from previous estimations, identify the cause and take necessary steps for adjustment



Capital Budgeting

Some Important Concepts

Capital Budgeting: Some Important Concepts

Evaluate Projects Based on Incremental Cash Flows

Projects should be evaluated based on the incremental cash flows. Primary forms of cash flows:

- **Initial Investment** → The investments made to establish the projects
 - Buying & installing machineries, setting up factories, building construction etc. working capital needed to support the project operations
- **Annual cash flows** → Net Cash Flows (in and outflows) generated during each year from the project operations (revenue net of operating & tax expenses)
- **Terminal Cash Flow** → At the end of the project, if any cash flow is generated by selling the assets used in the projects, as well as any working capital that is freed up

Capital Budgeting: Some Important Concepts

Project Classification

- **Stand Alone or Independent Projects** → **Each project is evaluated independently**
 - Projects are projects whose acceptance or rejection **DOES NOT** affect the acceptance of any other project
 - More than one project can be accepted
 - Decision on one project does not block another
- **Mutually Exclusive Projects** → **Maximum of only one project can be selected**
 - Selection of one project eliminates the other projects
 - Sometimes no projects would be selected (depending on the evaluation method)
 - All projects have negative NPV or all projects have IRR below Discount Rate (r)

Capital Budgeting: Some Important Concepts

Sunk Cost

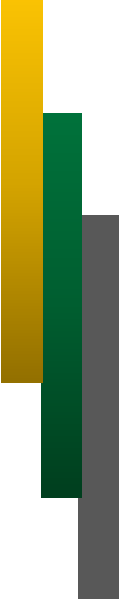
- A sunk cost is a cost that has already been incurred and cannot be recovered, regardless of whether a project is accepted or rejected
 - Sunk costs should be ignored in capital budgeting decisions
- A company spends BDT 5MN on market research to identify potential new products. Now it has 3 options of investment. This 5MN is sunk cost.

Capital Budgeting: Some Important Concepts



Discount Rate

- The appropriate discount rate for Project Evaluation is
 - **Weighted Average Cost of Capital (WACC)**
 - We'll discuss these in details in the next chapter



Capital Budgeting Techniques

Payback Period & Discounted Payback Period

Capital Budgeting Techniques: PBP & DPBP

Payback Period (PBP)

The time required to recover the initial investment from a project's cash flows

- **How quickly do we get our money back**
 - **The quicker the better**

PBP Decision Rule	Stand Alone (Independent) Project	Mutually Exclusive Project
Compare calculated PBP with our target time line for acceptance/ rejection decisions	Accept all projects with PBP less than target timeline	Only Accept the project with the shortest PBP, given that its less than the target timeline
	Reject Any project with PBP greater than the target time line	Reject all other projects

In case, no target timeline is mentioned, simply compare PBP with the project life time

Capital Budgeting Techniques: PBP & DPBP

Example Math: Grameen Phone (GP) is considering two mutually exclusive projects. Both projects would incur an initial investment of BDT 5,000. The Weighted Average Cost of Capital (WACC) for the firm is 11%

Year	Project A	Project B
1	3,000	1,000
2	2,500	3,000
3	1,500	5,000

Evaluate the projects based on:

- **Payback Period**
- **Discounted Payback Period**
- **NPV**
- **IRR**
- **Profitability Index**

Capital Budgeting Techniques: PBP & DPBP

Solve: How long each project takes to recover their respective initial investments:

Year	Project A Cash Flow	Project A Cumulative Cash Flow		Project B Cash Flow	Project B Cumulative Cash Flow	
0	-5,000		-5,000	-5,000		-5,000
1	3,000	$(-5,000+3,000)$	-2,000	1,000	$(-5,000+1,000)$	-4,000
2	2,500	$(-2,000+2,500)$	500	3,000	$(-4,000+3,000)$	-1,000
3	1,500	$(500+1,500)$	2,000	5,000	$(-1,000+5,000)$	4,000

PBP : From the Cumulative Cash Flow (CCF), we can see that it took project A more than 1 but less than 2 years to recover the initial investment (zero CCF). While it took project B more than 2 but less than 3 years to recover the initial investment (zero CCF). Thus we can start our calculations by considering 1 year as base for project A and 2 years and base for project B, and then add the fractional time period to reach PBP of the projects

Capital Budgeting Techniques: PBP & DPBP

$$PBP = \text{Base Years} + \left[\frac{\text{Cash Flow Required to Reach Zero CCF}}{\text{CF Generated During the Fractional Year}} \right]$$

$$PBP_A = 1 \text{ Year} + \left[\frac{2,000}{2,500} \text{ Year} \right] = 1.80 \text{ Years}$$

$$PBP_B = 2 \text{ Year} + \left[\frac{1,000}{5,000} \text{ Year} \right] = 2.20 \text{ Years}$$

For the two mutually exclusive projects, as per PBP criteria, we should accept project A & reject project B

Capital Budgeting Techniques: PBP & DPBP

Payback Period (PBP): Two Major Weaknesses

While a widely used method, PBP has 2 major weaknesses

- PBP **does not consider time value of money**, a core principles of Finance
- PBP method does not consider any cash flow generated after the initial investment has been recovered
 - Even if a project does tremendously well in its later phase, it may be ignored in PBP

Payback Period (PBP): Why It is so Widely Used

Despite its weaknesses, it is a widely used method, Primarily because of its simplicity

- Companies have a lot of small projects to evaluate, and PBP is used to streamline decision making process for these smaller projects

Capital Budgeting Techniques: PBP & DPBP

Discounted Payback Period (DPBP)

Discounted pay back period tackles one of the weaknesses of PBP and adjusts the cash flows for time value of money

- **How quickly do we get our money back considering time value of money**
 - **The quicker the better**

PBP Decision Rule	Stand Alone (Independent) Project	Mutually Exclusive Project
Compare calculated DPBP with our target time line for acceptance/rejection decisions	Accept all projects with DPBP less than target timeline	Only accept the project with the shortest DPBP, given that it's less than the target timeline
	Reject Any project with DPBP greater than the target time line	Reject all other projects

In case, no target timeline is mentioned, simply compare DPBP with the project life time

Capital Budgeting Techniques: PBP & DPBP

Solve: Considering a 11% discount rate (WACC) for our present value calculations

Year	Project A CF	Project A DCF	Project A DCCF	Project B CF	Project B DCF	Project B DCCF
0	-5,000	-5,000	-5,000	-5,000	-5,000	-5,000
1	3,000	2,703	-2,297	1,000	901	-4,099
2	2,500	2,029	-268	3,000	2,435	-1,664
3	1,500	1,097	829	5,000	3,656	1,992

Some Sample of Discounted Cash Flow (DCF) Calculations/Workings

$$Project A DCF Y1 = \left[\frac{3,000}{(1 + 11\%)^1} \right] = 2,703 \quad Project B DCF Y3 = \left[\frac{5,000}{(1 + 11\%)^3} \right] = 1,992$$

Capital Budgeting Techniques: PBP & DPBP

DPBP: From the DCCF, we can see that it took more than 2 but less than 3 years for both projects to recover their respective initial investments (zero CCF). Thus we can start our calculations by considering 2 year as base for both projects and then add the fractional time period to reach DPBP of the respective projects

$$DPBP = \text{Base Years} + \left[\frac{\text{DCF Required to Reach Zero DCCF}}{\text{DCF Generated During the Fractional Year}} \right]$$

$$DPBP_A = 2 \text{ Year} + \left[\frac{268}{1,097} \text{ Year} \right] = 2.24 \text{ Years}$$

$$DPBP_B = 2 \text{ Year} + \left[\frac{1,664}{3,656} \text{ Year} \right] = 2.46 \text{ Years}$$

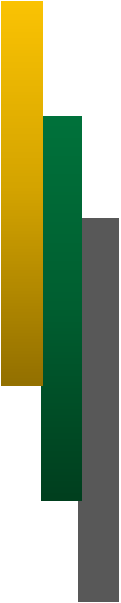
For the two mutually exclusive projects, as per DPBP criteria, we should accept project A & reject project B

Capital Budgeting Techniques: PBP & DPBP

Scenarios where PBP/DPBP May Become the Most Appropriate Method

Some unique scenarios may be best suited to be evaluated by PBP/DPBP

- For **fast changing industries** like the tech industry
 - A revolution in the industry may render existing products obsolete
- In cases of **political or economic uncertainty**
 - where life of the business may not be guaranteed
- In such unique scenarios, firms may look for safety and PBP/DPBP may become the most appropriate method of evaluation
- **Government projects** are not focused towards maximizing value
 - If the government project earns back its money quickly, the money can be reinvested faster in a new public project → Thus for government projects, PBP most suitable



Capital Budgeting Techniques

NPV, IRR & PI

Capital Budgeting Techniques: NPV

Net Present Value (NPV)

NPV is the sum of the present value of all future expected cash flows, net off against initial cash flow, which is commonly the initial investment

- **NPV tells us how much value a project adds today**
- **NPV is the best method for project evaluation → Solves both problems of PBP**
 - **If there is a conflict of decision between different methods, rely on NPV for decisions**

NPV Decision Rule	Stand Alone (Independent) Project	Mutually Exclusive Project
Projects with $NPV \geq 0$, is considered to be good	Accept all projects with $NPV \geq 0$	Only Accept the project with the highest NPV, given that its not negative
	Reject all negative NPV projects	Reject all other projects

Capital Budgeting Techniques: NPV

Going back to the previous math, two mutually exclusive projects with 5,000 BDT initial cost and WACC of 11% and the following cash flows:

Year	Project A	Project B
1	3,000	1,000
2	2,500	3,000
3	1,500	5,000

Solve: Discount all future CF using WACC and net off with initial investment

$$NPV = \left[\frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} \dots \dots \dots + \frac{CF_n}{(1+r)^n} \right] + CF_0$$

Capital Budgeting Techniques: NPV

$$NPV_A = \left[\frac{3,000}{(1 + 11\%)^1} + \frac{2,500}{(1 + 11\%)^2} + \frac{1,500}{(1 + 11\%)^3} \right] - 5,000 = 829$$

$$NPV_B = \left[\frac{1,000}{(1 + 11\%)^1} + \frac{3,000}{(1 + 11\%)^2} + \frac{5,000}{(1 + 11\%)^3} \right] - 5,000 = 1,992$$

For the two mutually exclusive projects, as per NPV criteria,
we should accept project B & reject project A

Capital Budgeting Techniques: IRR

Internal Rate of Return (IRR)

IRR is the discount rate for which NPV of the project would be zero

- Also tells us the annual return the project actually generating
- We compare it against the cost of fund (WACC)
 - To evaluate if the project is generating higher return compared to its cost of fund

IRR Decision Rule	Stand Alone (Independent) Project	Mutually Exclusive Project
Projects with $IRR \geq WACC$, is considered to be good	Accept all projects with $IRR \geq WACC$	Only Accept the project with the highest IRR, given that its not lower than WACC
	Reject all projects with $IRR < WACC$	Reject all other projects

Capital Budgeting Techniques: IRR

Going back to the previous math, two mutually exclusive projects with 5,000 BDT initial cost and WACC of 11% and the following cash flows:

Year	Project A	Project B
1	3,000	1,000
2	2,500	3,000
3	1,500	5,000

Solve: Calculate the IRR through trial error method

- I. Assume a random lower discount rate that would result in positive NPV
- II. Assume a higher discount rate that would result in a negative NPV
- III. Use IRR equation to identify the IRR that falls in between the two assumed rates
 - Note that, this is an approximate IRR, not a perfect IRR

Capital Budgeting Techniques: IRR

IRR for Project A

Let's assume the lower rate (L) is 11%

$$NPV_{AL} = \left[\frac{3,000}{(1 + 11\%)^1} + \frac{2,500}{(1 + 11\%)^2} + \frac{1,500}{(1 + 11\%)^3} \right] - 5,000 = 829$$

We can also skip this calculation at 11% if we want, as we've already calculated this back in our NPV part

Let's assume the higher rate (H) is 30%

$$NPV_{AH} = \left[\frac{3,000}{(1 + 30\%)^1} + \frac{2,500}{(1 + 30\%)^2} + \frac{1,500}{(1 + 30\%)^3} \right] - 5,000 = -530$$

$$IRR_A = L + \left[\frac{NPV_{AL}}{NPV_{AL} - NPV_{AH}} * (H - L) \right] = 11\% + \left[\frac{829}{829 - (-530)} * (30\% - 11\%) \right] = 22.59\%$$

Capital Budgeting Techniques: IRR



IRR for Project B

Let's assume the lower rate (L) is 11%

$$NPV_{BL} = \left[\frac{1,000}{(1 + 11\%)^1} + \frac{3,000}{(1 + 11\%)^2} + \frac{5,000}{(1 + 11\%)^3} \right] - 5,000 = 1,992$$

We can also skip this calculation at 11% if we want, as we've already calculated this back in our NPV part

Let's assume the higher rate (H) is 30%

$$NPV_{BH} = \left[\frac{1,000}{(1 + 30\%)^1} + \frac{3,000}{(1 + 30\%)^2} + \frac{5,000}{(1 + 30\%)^3} \right] - 5,000 = -180$$

$$IRR_B = L + \left[\frac{NPV_{BL}}{NPV_{BL} - NPV_{BH}} * (H - L) \right] = 11\% + \left[\frac{1,992}{1,992 - (-180)} * (30\% - 11\%) \right] = 28.43\%$$

For the two mutually exclusive projects, as per IRR criteria, we should accept project B & reject project A

Capital Budgeting Techniques: IRR

Internal Rate of Return (IRR): Weaknesses

While a very strong method of evaluation, IRR does have some weaknesses

- In case of non-conventional cash flows there might be multiple IRRs or no IRR at all
 - **Non-conventional cash flow pattern:** When the cash flows change between inflows (positive) and outflows (negative) more than once
- IRR assumes that the cash inflows generated by the project are reinvested at the IRR itself
 - In reality the cash flows maybe reinvested at higher/lower than the IRR
- Sometimes overestimates the results of projects with smaller initial investment

Capital Budgeting Techniques: PI

Profitability Index (PI)

The ratio of the present value of future cash inflows to the initial investment of a project

- **PI shows how much value is created per taka invested**

$$PI = \frac{\sum \text{Present Value of all Future Cash Flows}}{\text{Initial Investment}}$$

PI Decision Rule	Stand Alone (Independent) Project	Mutually Exclusive Project
Projects with $PI \geq 1$, is considered to be good	Accept all projects with $PI \geq 1$	Only Accept the project with the highest PI, given that its not lower than 1
	Reject all projects with $PI < 1$	Reject all other projects

Capital Budgeting Techniques: PI

Going back to the previous math, two mutually exclusive projects with 5,000 BDT initial cost and WACC of 11% and the following cash flows:

Year	Project A	Project B
1	3,000	1,000
2	2,500	3,000
3	1,500	5,000

Solve: Discount all future CF using WACC and divide by initial investment

$$PI_A = \frac{\left[\frac{3,000}{(1 + 11\%)^1} + \frac{2,500}{(1 + 11\%)^2} + \frac{1,500}{(1 + 11\%)^3} \right]}{5,000} = \frac{5,829}{5,000} = 1.17$$

Since we already know the NPV of project A to be 829, we can add back 5,000 initial investment to the NPV 829 to calculate 5,829 in the numerator of the PI equation to skip the lengthy calculation

Capital Budgeting Techniques: PI

$$PI_B = \frac{\left[\frac{1,000}{(1 + 11\%)^1} + \frac{3,000}{(1 + 11\%)^2} + \frac{5,000}{(1 + 11\%)^3} \right]}{5,000} = \frac{6,992}{5,000} = 1.40$$

Since we already know the NPV of project B to be 1,992, we can add back 5,000 initial investment to the NPV 1992 to calculate 6,992 in the numerator of the PI equation to skip the lengthy calculation

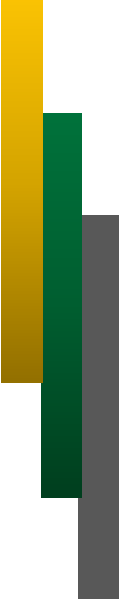
**For the two mutually exclusive projects, as per PI criteria,
we should accept project B & reject project A**

Capital Budgeting Techniques: PI

Profitability Index (PI): Weaknesses

PI also have some weaknesses

- Overestimates the results of projects with smaller initial investment
- Has to be compared to another project for interpretation if it is good or not



Capital Budgeting Techniques

Capital Rationing

Capital Budgeting Techniques: Capital Rationing

Capital Rationing

Choose the combination of projects that combinedly provides the highest value

Example Math: Imagine you have a total of BDT 100MN fund available and have the following independent projects that you can take on:

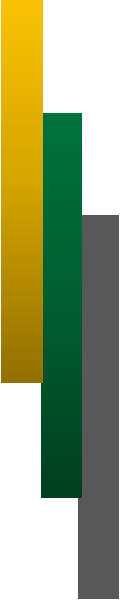
Project	Initial Investment	NPV of Project
A	100MN	19MN
B	50MN	11MN
C	20MN	5MN
D	50MN	10MN
E	20MN	4MN

Capital Budgeting Techniques: Capital Rationing

Solve: Find the combination with the highest NPV

Possible Combinations	Combined Initial Investment	Combined NPV
A Alone	100MN	19MN
B & D	$(50+50) = 100\text{MN}$	$(11+10) = 21\text{MN}$
B, C & E	$(50+20+20) = 90\text{MN}$	$(11+5+4) = 20\text{MN}$
C, D & E	$(20+50+20) = 90\text{MN}$	$(5+10+4) = 19\text{MN}$

Under the consideration of Capital Rationing, Combination of Project B & D provides the highest combined NPV. Thus Projects B & D should be accepted and the other three projects should be rejected



Capital Budgeting Techniques

Assignment

Capital Budgeting Techniques: Assignment

Example Math: Grameen Phone (GP) is considering two mutually exclusive projects. Both projects would incur an initial investment of BDT 5,000. The Weighted Average Cost of Capital (WACC) for the firm is the last digit of your ID.

For example if your ID is 1085, consider the WACC to be 5%. If the last digit of your ID is 0, in that case assume a 10% discount rate

Year	Project A	Project B
1	3,000	1,000
2	2,500	3,000
3	1,500	5,000

Evaluate the projects based on:

- **Payback Period**
- **Discounted Payback Period**
- **NPV**
- **IRR**
- **Profitability Index**

Capital Budgeting

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